

Review Questions

Digital Signal Processing | Chapters 1–3

Master-level Course

Part I. Conceptual Questions

Signals

1. Explain the differences between:
 - Analog and digital signals.
 - Continuous-time and discrete-time signals.
 - Energy and power signals.
2. Why is digital signal processing preferred over analog signal processing in modern applications?
3. Give three real-world applications of DSP.

Sampling & PCM

4. Explain the three main steps of Analog-to-Digital Conversion (ADC).
5. What is Pulse Code Modulation (PCM)?
6. Why is quantization necessary?
7. What is quantization noise?
8. State the Nyquist Sampling Theorem.
9. Explain aliasing using the frequency-domain perspective.
10. Why must an anti-aliasing filter be used before sampling?
11. What is the relationship between analog frequency and digital frequency?
12. Given a sampling frequency f_s , what is the maximum frequency that can be represented without aliasing?

Systems

13. What are the properties of an LTI system?
14. How can impulse response completely characterize an LTI system?
15. What is convolution? Why is it important in DSP?

z-Transform

16. Why do we need the z-transform?
17. Compare Fourier Transform and z-transform.
18. What is the Region of Convergence (ROC)?
19. How can ROC determine:
 - stability?
 - causality?

20. What is the physical meaning of poles and zeros?

21. What is the transfer function?

22. Why is the following relation important? $H(e^{j\omega}) = H(z) \big|_{z=e^{j\omega}}$

Part II. Short Problems

Problem 1

- . A signal has a bandwidth of 3 kHz.
 - What is the minimum sampling frequency?

Problem 2

- . If $f_s = 8000$ Hz, what is the Nyquist frequency?

Problem 3

- . Convert $f = 1000$ Hz into digital frequency when $f_s = 8000$ Hz.

Problem 4

- . Determine whether the following systems are:
 - Linear
 - Time-invariant
 - Causal
 - Stable

Problem 5

- . Sketch the pole-zero plot and determine whether the system is stable.

Part II-B. Mini Computational Questions (≈ 2 min each)

Quick calculations to reinforce sampling and frequency concepts. Each question can be solved in under 2 minutes.

MCQ1 — Sampling Basics ($f_s = 8$ kHz)

Given: $f_s = 8000$ Hz

- a. What is the **Nyquist frequency**?
- b. Calculate the **digital frequency** ω of an analog frequency $f = 2000$ Hz.
- c. An analog tone at $f = 6000$ Hz is sampled at $f_s = 8000$ Hz. What is the **aliasing frequency** that appears in the digital signal?

MCQ2 — Anti-Aliasing

- a. A signal contains components at 500 Hz, 3.5 kHz, and 5 kHz. It is sampled at $f_s = 8$ kHz. Which components alias? Where do they appear?
- b. What is the minimum sampling rate needed to capture a signal with components up to 4.5 kHz?

MCQ3 — Digital Frequency Conversion

Given: $f_s = 16$ kHz

- a. Convert $f = 4$ kHz to digital frequency ω (in radians/sample).
- b. Convert digital frequency $\omega = \pi/4$ back to analog frequency in Hz.

Part III. Discussion

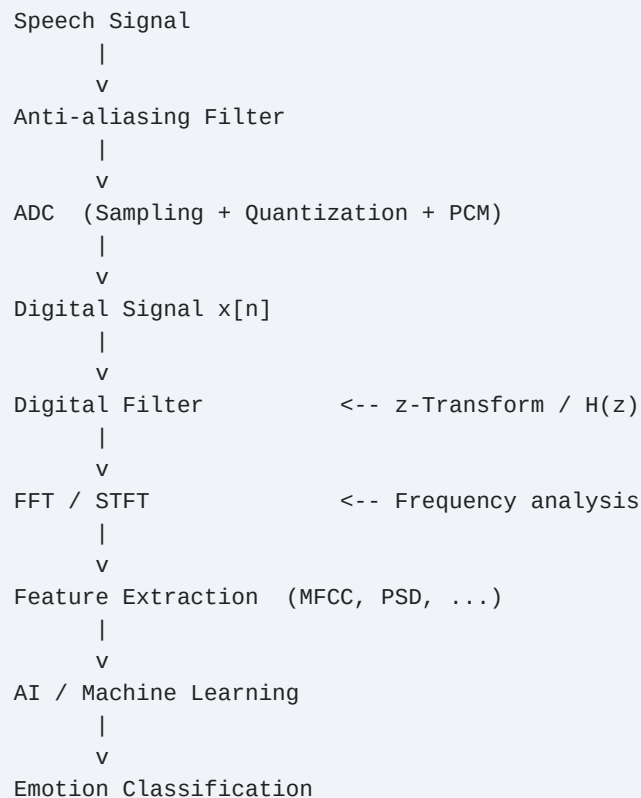
Strongly recommended for Master-level students.

1. Why is Fourier Transform alone insufficient for analyzing digital systems?
2. Why is the z-transform considered the most important mathematical tool in DSP?
3. If you were designing a speech enhancement system, where would the following appear?
 - Sampling
 - PCM
 - z-transform
 - Digital Filter

Big Picture Question

Explain the **complete processing pipeline** from an analog speech signal to a **speech emotion recognition system**. Indicate where each component is applied:

- Sampling
- PCM
- z-Transform
- FFT / STFT
- Digital Filtering
- Feature Extraction
- AI / Machine Learning



Expected Learning Outcomes

After completing this review, students should be able to:

- ✓ Distinguish analog vs. digital and energy vs. power signals
- ✓ Apply the Nyquist Theorem and explain aliasing
- ✓ Perform basic digital frequency calculations
- ✓ Identify properties of LTI systems (linearity, time-invariance, causality, stability)
- ✓ Compute and interpret convolution
- ✓ Define ROC and explain its role in stability and causality
- ✓ Relate the z-transform to the frequency response $H(e^{j\omega})$
- ✓ Describe the end-to-end DSP pipeline from ADC to AI